

Sharp partial-pivoting growth for unequal bandwidths

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Independently reviewed resolution. The complete proof was checked against the current canonical target IE-13. The recovered provisional identifiers are confirmed. Authorship is recorded at the submitter's request; substantial AI assistance and reconstruction are disclosed in the submission record. The dated independent agent review records the subsequent checks and supersedes original draft remarks about pending review. This is not external human peer review or formal certification.

Reconstructed candidate proof draft. Independent mathematical review and source/priority verification are required. The IE identifiers are provisional labels recovered from earlier notes, not verified current repository identifiers.

Abstract

For lower bandwidth $p \geq 1$ and upper bandwidth $q \geq 0$, the proposed optimal dimension-independent GEPP growth factor is h_{p+q} , where $h_t = 0$ for $t \leq 0$ and $h_t = 1 + \sum_{r=1}^p h_{t-r}$ for $t \geq 1$. A sorted-front argument gives an upper bound, and an explicit nonsingular real band matrix attains it. The proof covers fixed column order, real and complex matrices, and arbitrary permitted pivot ties.

1 Statement

Let $A \in \mathbb{R}^{n \times n}$ or $\mathbb{C}^{n \times n}$ be nonsingular, with $a_{ij} = 0$ whenever $i > j + p$ or $j > i + q$. Normalize $\max |a_{ij}| = 1$. Growth includes all active Schur-complement entries, not merely U . Define $G(p, q)$ as the supremum over permitted orders and pivot paths.

Theorem 1. $G(0, q) = 1$. For $p \geq 1$,

$$h_t = 0 \quad (t \leq 0), \quad h_t = 1 + \sum_{r=1}^p h_{t-r} \quad (t \geq 1), \quad \boxed{G(p, q) = h_{p+q}}.$$

A real matrix of order $2p + q + 1$ attains the bound.

The recurrence gives

$$\begin{aligned} G(p, 0) &= 2^{p-1}, \\ G(p, q) &= 2^{p+q-1} - (q-1)2^{q-2} \quad (1 \leq q \leq p+1), \\ G(1, q) &= q+1, \quad G(2, q) = F_{q+4} - 1, \end{aligned}$$

where $F_0 = 0, F_1 = 1$. In particular the equal-bandwidth specialization is $2^{2p-1} - (p-1)2^{p-2}$. These identities follow by induction.

2 Sorted-front lemma

Suppose a front has $p + 1$ target-column magnitudes $y_1 \geq \dots \geq y_{p+1} \geq 0$. Remove the row with target magnitude y_a and update the others with multipliers of modulus at most one. The sorted survivors are bounded componentwise by

$$(y_1 + y_2, y_1 + y_3, \dots, y_1 + y_{p+1}).$$

Indeed, the triangle inequality bounds each survivor by $y_a + y_i$. For $i < a$, $y_a + y_i \leq y_1 + y_{i+1}$, since $y_i - y_{i+1} \leq y_1 - y_a$. For $i \geq a$, the ordered bound is $y_a + y_{i+1} \leq y_1 + y_{i+1}$. Sorting and appending a fresh value are monotone. Thus always eliminating the largest target-column magnitude gives a componentwise upper envelope, even though GEPP actually chooses by another column.

Start with p zeros, append a fresh 1, and perform this envelope update. After one update all survivors equal 1. Subsequently the ordered state evolves by

$$(x_1, \dots, x_p) \mapsto (x_1 + x_2, \dots, x_1 + x_p, x_1 + 1).$$

After $t \geq 1$ updates its i th entry is

$$x_i^{(t)} = 1 + \sum_{r=1}^{p-i+1} h_{t-r},$$

so its largest entry is h_t . Direct substitution proves this identity, including $p = 1$.

3 Upper bound

At the start of stage k , original rows with labels $i \geq k + p$ are unchanged by earlier stages: their first $k - 1$ original entries are zero. They have zero earlier multipliers and cannot be selected as nonzero earlier pivots. Thus there are at most p old surviving rows plus fresh original row $k + p$; later rows are untouched. Near the bottom, missing fresh rows can be padded by zeros.

Fix target column j and set $L = p + q$. For $j \geq L + 1$, all target entries in the front are zero until stage $k_0 = j - L$, when original row $j - q$ arrives. Exactly L updates occur from k_0 through $j - 1$. Each fresh target entry has modulus at most one, so the lemma bounds every value by h_L . For $j \leq L$, the initial p old values are bounded by the canonical state after one imaginary update. At most $j - 1$ actual updates precede elimination of column j , giving $h_j \leq h_L$. Recorded pivot-row entries and untouched entries satisfy the same bound. This covers every active entry. For $p = 0$ the matrix is upper triangular and all elimination multipliers vanish.

4 Attaining construction

For $p \geq 1$, put $j = p + q + 1$, $n = j + p$, and let L_0 be unit lower triangular with -1 on its first p subdiagonals. Use the original-row order

$$\sigma = (p + 1, 1, 2, \dots, p, p + 2, \dots, n), \quad (Pa)_i = a_{\sigma_i}, \quad \eta = 2^{-p}.$$

For $k < j$, define an upper-supported vector $u^{(k)}$ by

$$u_i^{(k)} = \begin{cases} 1 & i = 1, \ k \leq p + 1, \\ 2^{i-2} & 2 \leq i \leq k \leq p + 1, \\ 1 & i = k, \ k \geq p + 2, \\ 0 & \text{otherwise,} \end{cases} \quad A_{:,k} = \eta P^T L_0 u^{(k)}.$$

Set

$$A_{i,j} = \begin{cases} 1 & p+1 \leq i \leq n, \\ 0 & i \leq p, \end{cases} \quad A_{:,k} = e_k \quad (k > j).$$

For $k \leq p+1$, the first entry of $L_0 u^{(k)}$ is 1, entries $2, \dots, k$ are zero, and later entries are negatives of sums of subsets of $u^{(k)}$. Their moduli are at most $2^{k-1} \leq 2^p$. After row-label assignment, support lies in original rows $k, \dots, k+p$. For $k \geq p+2$ the same support follows from $L_0 e_k$. Hence these columns have lower bandwidth at most p and upper bandwidth zero, with entries bounded by one. The target column starts at $p+1 = j-q$ and ends at $n = j+p$. Later identity columns respect the band. The initial maximum is one.

In the first $j-1$ stages in row order σ , the factorization gives multipliers in $\{0, -1\}$ and nonzero pivots. After earlier eliminations, active column k is $L_{0,ik} U_{kk}$; the diagonal multiplier is one, so each chosen pivot is maximal in modulus. Thus these are admissible GEPP choices, not an extra preliminary permutation.

For the target column, $c = P A_{:,j}$ has $c_1 = 1$, $c_i = 0$ for $2 \leq i \leq p+1$, and $c_i = 1$ for $i \geq p+2$. Forward substitution gives

$$u_i = c_i + \sum_{r=1}^p u_{i-r}, \quad u_i = 0 \quad (i \leq 0).$$

Induction gives $u_1 = 1$ and $u_i = h_{i-1}$ for $i \geq 2$. Therefore the active (j, j) entry at stage j equals h_{p+q} . The leading $j \times j$ block has nonzero pivots, and later columns are identity columns; the matrix is block lower triangular with nonsingular diagonal blocks. This proves attainment.

5 Checks and source status

The standard-library exact verifier checks 60 attaining matrices ($1 \leq p \leq 6$, $0 \leq q \leq 9$), their support, pivot admissibility, growth, and nonsingularity, plus recurrence identities. These finite checks are not a formal proof of all orders.

Recovered target description: *Sharp growth for unequal lower and upper bandwidths*, provisionally IE-13 in OpenProblemsInNLA. The current repository statement could not be retrieved during reconstruction. The result is therefore submitted for review under the precise hypotheses above, not asserted to resolve a source statement whose scope has been independently verified. Repository supplied for the task:¹

¹<https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/linear-systems-and-elimination>.